

A brief example on conditional expectations, conditional probabilities, and algebras

ECO403/503–Introduction to Contract Theory and Auctions

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The purpose of this brief note is to provide you with a simple example, intended for study and discussion, about calculating conditional expectations, conditional probabilities, and how these behave under different levels of information, as modeled with the use of algebras.

1 Tossing a fair coin two times

Consider the experiment of an player tossing a fair coin two times. The probability space modelling this situation is given by $\Omega = \{HH, HT, TH, TT\}$, algebra $\mathcal{F} = 2^\Omega$ and prior distribution $\mu(\omega) = 1/4$ for all $\omega \in \Omega$.

2 A sequential view of the experiment and the arrival of new information

Suppose that we carry out the experiment described in section 1 in a sequential manner. There are three times:

Time $t = 0$: This time represents the moment before the coins are tossed. At this time, the **only** meaningful events that can be talked about or the only information that the individual has, before tossing the coins, is represented by the following algebra: $\mathcal{F}_0 = \{\emptyset, \Omega\} \subset \mathcal{F}$. The interpretation of the previous algebra is that at time $t = 0$, all outcomes are deemed possible by the player throwing the coin. Notice that the algebra $\mathcal{F}_0$ can also be understood as arising from the following **information structure**: at time $t = 0$, the player has a type space $T_0 = t_0$ and a signalling function $\tau_0(\omega) = t_0$ for all $\omega \in \Omega$; the interpretation is that at time $t = 0$, before tossing the coins, the player is not informed at all about the outcome of the coin tosses, which leads to the player deeming possible all states, creating her initial partition $\mathcal{P}_0 = \{\Omega\}$, which generates the algebra $\mathcal{F}_0$.

Time $t = 1$: This time represents the moment after the first coin toss has been done. At this time, the player knows that the events in the following algebra could happen: $\mathcal{F}_1 = \{\emptyset, \Omega, \{HH, HT\}, \{TT, TH\}\}$. Notice that $\mathcal{F}_0 \subset \mathcal{F}_1$, so $\mathcal{F}_1$ contains **more information** than $\mathcal{F}_0$. Another way of seeing this is again using the following information structure: at time $t = 1$, the player has a type space $T_1 = \{t_{1a}, t_{1b}\}$, and a signalling function $\tau_1(HH) = \tau_1(HT) = t_{1a}$, $\tau_1(TT) = \tau_1(TH) = t_{1b}$. The previous information structure generates the partition $\mathcal{P}_1 = \{\{HH, HT\}, \{TT, TH\}\}$, which of course generates the algebra $\mathcal{F}_1$.

Time $t = 2$: At this time, the coin has been tossed twice, and thus the player has the information, represented by the algebra $\mathcal{F}_2$, that is generated by the partition $\mathcal{P}_2 = \{\{HH\}, \{HT\}, \{TH\}, \{TT\}\}$. Notice that the previous partition is generated by a fully informative signal structure, consisting of four types and a signalling function mapping each type to a different state. For the sake of completeness, I write in full the algebra $\mathcal{F}_2$ below:

$$\mathcal{F}_2 = \{\emptyset, \Omega, \{HH\}, \{HT\}, \{TH\}, \{TT\}, \{HH, HT\}, \{HH, TH\}, \{HH, TT\}, \{HT, TH\}, \{HT, TT\}, \\ \{TH, TT\}, \{HH, HT, TH\}, \{HH, HT, TT\}, \{HT, TH, TT\}, \{HH, TH, TT\}\}.$$

Notice that $\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2$.

3 An illuminating random variable and its conditional expectation at different levels of information

Consider the following random variable $X : \Omega \rightarrow \mathbb{R}$.

$$X(\omega) = \begin{cases} 0 & \omega = TT, \\ 1 & \omega \in \{HT, TH\}, \\ 2 & \omega = HH. \end{cases}$$

The expectation $E[X|\mathcal{F}_0](\omega) = (1/4)(0 + 1 + 1 + 2) = 1$, for all states $\omega \in \Omega$. Notice that this expectation is computed at the *coarsest* algebra $\mathcal{F}_0$, which is when the player has less information. Interestingly, the previous conditional expectation with the coarsest possible information is what you are used to call $E[X]$, i.e., the expectation of X . Intuitively, the most we can say about the random variable X with the coarsest possible information is $E[X]$.

Now, what about $E[X|\mathcal{F}_1]$? Here more interesting things start to happen since the player now has access to what happened in the first throw of the coin.

$$E[X|\mathcal{F}_1](\omega) = \begin{cases} \frac{(1/4)(1+2)}{1/2} = 3/2 & \omega \in \{HH, HT\}, \\ \frac{(1/4)(0+1)}{1/2} = 1/2 & \omega \in \{TT, TH\}. \end{cases}$$

Notice that the conditional expectation changes depending on the information that the player has. Intuitively, if the information is “the first throw was H ”, then the player updates her beliefs and computes the new expectation accordingly; if the information is “the first throw was T ”, the player updates her beliefs and computes the new expectation accordingly.

Finally, consider $E[X|\mathcal{F}_2]$. At time $t = 2$, the player has all the information about the results of the coin throws. Therefore, as you can check, the conditional expectation then is equal to

$$E[X|\mathcal{F}_2](\omega) = X(\omega) \quad \text{for all } \omega \in \Omega.$$

The interpretation of the above is that with the information contained in $\mathcal{F}_2$, the player knows for sure the value of the random variable $X(\omega)$ at all states of the world ω , and thus, the conditional expectation given this information equals the value of the random variable.